L8.4: The Gram-Schmidt process

Transcript Language

English

In this video we are going to talk about the Gram-Schmidt process. So, what is the Gram-Schmidt

process? Let us look at this process in a nutshell.

So, what the Gram-Schmidt process does is that in an inner product space it takes any

basis x1, x2, xn and it produces, we use the Gram-Schmidt process and it produces an orthonormal

basis v1, v2, vn. So, let us recall before we start going ahead what is a basis. So, a basis is a linearly independent set which is also spanning.

So, in other words every element of your vector space or in this case for inner product space

can be written as a unique linear combination of these vectors x1, x2, xn. So, this is true

for any vector space that we can produce a basis and we have seen how to do that, but

suppose now that you have an inner product space. So, you have not only a vector space

but on that vector space you also have an inner product.

Then you can use the Gram-Schmidt process to get from these x1, x2, xn, which is a basis,

an orthonormal basis, which is, just to recall that each of these vectors are mutually orthogonal,

that means the inner product of vi, vj is 0 if i is not equal to j, and each of these

vectors have norm, that means the inner product of vi with itself so vi comma vi inner product

is 1. So, that is an orthonormal basis.

And in the previous videos we have seen why orthonormal basis are useful, because you can express various phenomenon in very nice ways. In particular you can write down the

linear combination of any vector. The constants that appear in the linear combination or the

coefficients that appear in the linear combination very nicely, in terms of the inner product,

and further they represent something called the projection which we saw in the previous

video.

So, the project can be written or is the inner product of the given vector with some of these

vectors depending on where you want to project and that is one of the, I mean you have to

divide by something and so on. So, there was a formula that we found out last time, but

that formula was particularly easy if you are, first of all there is a formula if have

an inner product, if you have orthogonal basis and if it is orthonormal the formula is particularly,

you do not even have to divide by anything.

So, even in this video we are going to use this idea of projection and that is the main sort of key tool in the Gram-Schmidt process.

So, let us do an example and the intuition behind that, and study the intuition behind

the Gram-Schmidt process. So, let us look at this basis for R3, 1, 2, 2; minus 1, 0,

2 and 0, 0, 1. So, first of all you have to convince yourself that this is a basis. So,

recall that we know how to get basis for Rn by, if you have given n vectors how to check

if it is a basis, you just take those n vectors, put them into a metrics as the columns or

it is okay even as rows, and you compute the determinant. So, if the determinant is non

0, then it will be a basis and you can check that indeed this is a basis. So, I will leave

that checking to you.

So, the question is can we use this to obtain an orthonormal basis for R3? So, now we want

to convert from this basis to another basis which is an orthonormal basis. Now, as we

have mentioned earlier in the convention here is that if no inner product is specifically mentioned then that means you are looking at the standard inner product, the dot product

of R3. So, this is an orthonormal basis with respect to the dot product. So, that is what

we want to obtain, an orthonormal basis with respect to the dot product.

So, we start by taking 1, 2, 2, well if you want to get something, so let us start with

the first vector. Now we want a vector which is orthogonal to v1, so let us first try to

get an orthogonal basis. So, that means we want to get v1, v2, v 3, so that they are

mutually orthogonal. Then once we have an orthogonal basis we know how to go from there

to orthonormal, by dividing by each vector by its norm.

So, let v1 be 1, 2, 2; we want a vector which is orthogonal v1. So, last time we saw that

vectors which are orthogonal to v1 are exactly those which are in, what we call v1 perp or

in general if you have a subspace w, then vectors which are orthogonal to all vectors

in w are exactly, that set which happens to be a subspace is exactly the set or the subset w perp. So, we want a vector which is in v1 perp or span of v1 perp. So, we use the projection

P v1 to v1. How do we use this?

Well, here is how we use it. define v2 to be, you take the second basis vector minus

1, 0, 2, so you take this vector minus 1, 0, 2 and you subtract the projection of minus

1, 0, 2 onto v1 from minus 1, 0, 2, that is what you do, and this is a standard computation

which is something we have done in the previous video namely P1 of minus 1, 0, 2 is some scalar

multiple of v1 and what is that scalar multiple? It is the inner product of minus 1, 0, 2 with

1, 2, 2 divided by the norm square of 1, 2, 2.

So, if you compute this you get, the numerator is minus 1, plus 4 so 3 and the denominator

is 1 plus 4 plus 4, so 9. So, 3 by 9 is 1 by 3. So I get minus 1, 0, 2; minus 1 by 3

times 1, 2, 2 and then if you do the subtraction this is what you get. Minus 4 thirds, minus

2 thirds, 4 thirds. So, first of all why did we do this? So, this was because remember

last time that we saw that W perp, so we have any subspace W, then you have this other subspace

called W perp which is all those vectors v, such that v comma.

Let me write this carefully, v, such that inner product of v and W is 0 for all w in

W. It was this set and then we saw it is a subspace because we saw that this is the null

space of Pw. So, what are we doing here? To obtain something in W perp, what we should

do is we should get something so that when we evaluate Pw on that then it gives us 0.

So, Pw of v equal 0 if an only if v belongs to W perp, this was what we saw.

And the point is if you look at the identity minus Pw so this is, you evaluate this on

v, if you evaluate this on v so you get v minus Pw of v and if you again take Pw of

this, so now you have already projected, after you apply Pw you have already projected in

W, so remember that Pw composed Pw was just Pw because the projection does not change

elements in W and that gives us that this is 0. This was something we studied.

Which means that if you take any element of this firm over here, this is inside W perp.

So, this is what we have seen and that is exactly what we are using here, we are using

that if you do minus 1, 0, 2; minus P v1 of minus 1, 0, 2 so you can rewrite this as 1

minus P v1, identity minus P v1 of minus 1, 0, 2 and this vector will be in v1 perp.

This is what the idea is for this computation, and that is why we do this computation, we

get v2 and this explanation below is a general method to see that v2 is in v1 perp, but you

can actually explicitly do this by taking this vector v2 and computing v2 comma v1,

I will encourage you to check that this is actually 0. So, let us move on.

So, now we have v1 and v2 which are orthogonal to each other, so now we want a vector which

is orthogonal to both v1 and v2, because remember we want to get an orthogonal basis. So in

other words we want this vector let us call it v3 to be in the span of v1, v2 perp. That is what we want, and how do we do this? When we again use the projection and this is exactly

the idea that we had in the previous slide. You take this span of v1 comma v2 as W and

you apply, 1 minus, I minus Pw on some vector. You apply this on some vector and then you

are going to get a vector which is perpendicular to v1 and v2.

The only thing is you have to be careful that this vector that you have should not be already

in the span of v1, v2 and that is where this basis comes in. So, the third vector is not

in the span of v1, v2; how do I know that? Because the span of v1, v2 is the same as

the span of the previous vectors that we had, because you see this 1, 2, 2 is just v1 and

minus 1, 0, 2 and v1, the span, is the same as the span of v1 and v2. So, that is another

thing you should check because they are linear combinations of each other.

So, we know that v3 or the third basis vector is not in the span of v1, v2 and that is how

we are going to use it. So, define v 3 to be 0, 0, 1; minus P v1 of 0, 0, 1; minus P v2 of 0, 0, 1. Now P W of 0, 0, 1 is exactly P v1 of 0, 0, 1 plus P v2 of 0, 0, 1. This

is something we saw in the previous video and now you can evaluate this. I will encourage

you to do that. It is a little bit of a computation and if you do it correctly, and I hope I have

done it correctly then I have got 2 by 9 comma minus 2 by 9 comma 1 by 3. I will encourage

you to check that.

And the point is, this is an orthogonal basis in orthogonal basis. And I do not even need

to check this because of what I did in the previous slide. I mean I gave you a general

theory for how to do this. So, this is an orthogonal basis. Now if you have not understood

why it is an orthogonal basis, I will encourage you again to check that this is an orthogonal

basis. So, check that v1, v2 is v1, v3 is v2, v3 is 0. If you have not completely followed

what I said in the previous slide.

So, in that case you will get that v1, v2 v3 is an orthogonal basis and now you divide each vector by its norm. I will encourage you to check what is the norm and then if

you do that you get the orthonormal basis 1 third, 2 third, 2 third; minus 2 third,

minus 1 third, 2 third and 2 third, minus 2 third, 1 third. You may remember this basis

from a previous video.

So, now let us do the Gram-Schmidt process. The intuition we have seen already, you start

with a basis and then you keep modifying the vectors in that basis. So, as to get an orthogonal

basis and further you will be dividing by the norm. So, as to make it orthonormal. Now

the Gram-Schmidt process kind of takes this orthogonal basis and divide by the norm in

one shot meaning it does not wait till the end to divide. It starts dividing right from

the start. The reason being it helps for efficiency, computational efficiency.

So, let us start with the basis x1, x2, xn and define the orthogonal basis v1, v2, vn

and the corresponding orthonormal basis w1, 2, w n as follows. So, we v1 is x1, w 1 is v1 by norm of v1. This is what we saw in the previous example. v2 is x2 minus x2 comma

w 1 inner product, times w1. So now why did not I, in the previous slides, in previous

example I took v2, x2 minus x2 comma v1,inner product and then divided by norm of v1 comma

v1 into v1.

So, the thing is if you already convert your basis vectors into the orthonormal form then

you need not divide by their norm because you know that the norm is 1, and this is what

I meant by saying that there is some computational efficiency. So, you do not have to keep dividing

by its norm. And then w 2 is v2 by norm of v2, so you have, v1 and v2 is an orthogonal

basis with span the same as x1, x2 and w1 and w2 is orthonormal basis which, I should

be careful, it is not a basis, so I take back the word basis.

These are linearly independent vectors, but which are orthogonal and w1 and w2 are orthonormal

vectors. So, they are automatically linearly independent. So, we will keep going. What is the general term? So, the i th term will be given there, suppose you have reached i

minus 1 at stage, if you have vi minus 1 and wi minus 1, all the vectors are still there.

So, vi is xi minus inner product of xi comma w1 times w1 minus inner product of xi comma

2 times w2 minus all the way up to inner product of xi comma wi minus 1 times wi minus 1, and

then wi is you normalize this vector.

Which means if you take vi and divide by its norm, and you keep going till the end and

in vn you have all the previous vectors defined till n minus 1 and then vn is xn minus inner

product xn comma w 1 times w 1 minus inner product xn comma w 2 times w 2, all the way

up to inner product xn comma w n minus 1 times w n minus 1, and then w n is vn by norm of

vn.

So, now the main point here is that if you take this basis w 1, w2, wn it is an orthonormal

basis. v1, v2, vn is an orthogonal basis and the reason this works, this is the main point, is that span of the set x1, x2, xi minus 1 is equal to the span of the set v1, v2, vi

minus 1 and which is again equal to the span of the set, w1, w2, wi minus 1. And because

these spans are the same we know that x i does not belong to this span and we can carry

on.

So, the main take-homes in this, maybe before I come here I will point out also that this

part here is exactly equal to P span of w1 through wi

minus 1 of xi. This is the projection of xi, on to the subspace, spanned by w1, w2, wi

minus 1, and this comment appear, tells you that this is as a result the same as the projection

of xi on the span of x1, x2, xi minus 1. So, really this entire thing is about projecting

carefully and as we know projections give you things which are orthogonal. This was

the main point of the previous projection's video.

So, with that brief comment I will go to the next and final slide which is what the main take-homes of this entire video. So, the main take-homes are that if you take any finite-dimensional

vector space with an inner product, so in other words a finite-dimensional inner product

space, then it has an orthonormal basis, this was not something obvious from the previous

videos that you can always get an orthonormal basis, but now not only can we get an orthonormal

basis we in fact have an explicit process. So, any basis can be changed to an orthonormal

basis using the Gram-Schmidt process.

So, how do you get an orthonormal basis? You start with any basis, so the assumption here

is that you can get a basis and indeed we have seen in previous videos how to get a

basis and, when I say previous videos, this is a few, maybe a week or so back or even

before that and then now once you get a basis you can change it to orthonormal basis by

using this Gram-Schmidt process.

So, the idea often is that when you have a, I mean when you are doing, using this kind of mathematics, typically you happen to know a basis, yeah, you will somehow be able to

construct some basis and then it is some algorithm, namely the Gram-Schmidt process, that will

allow you to actually get an orthonormal basis, and once you get an orthonormal basis expressing

any vector in terms of this orthogonal basis becomes very easy. That is what you have seen

in the previous videos. So, with this I guess I will end this video. Thank you.